

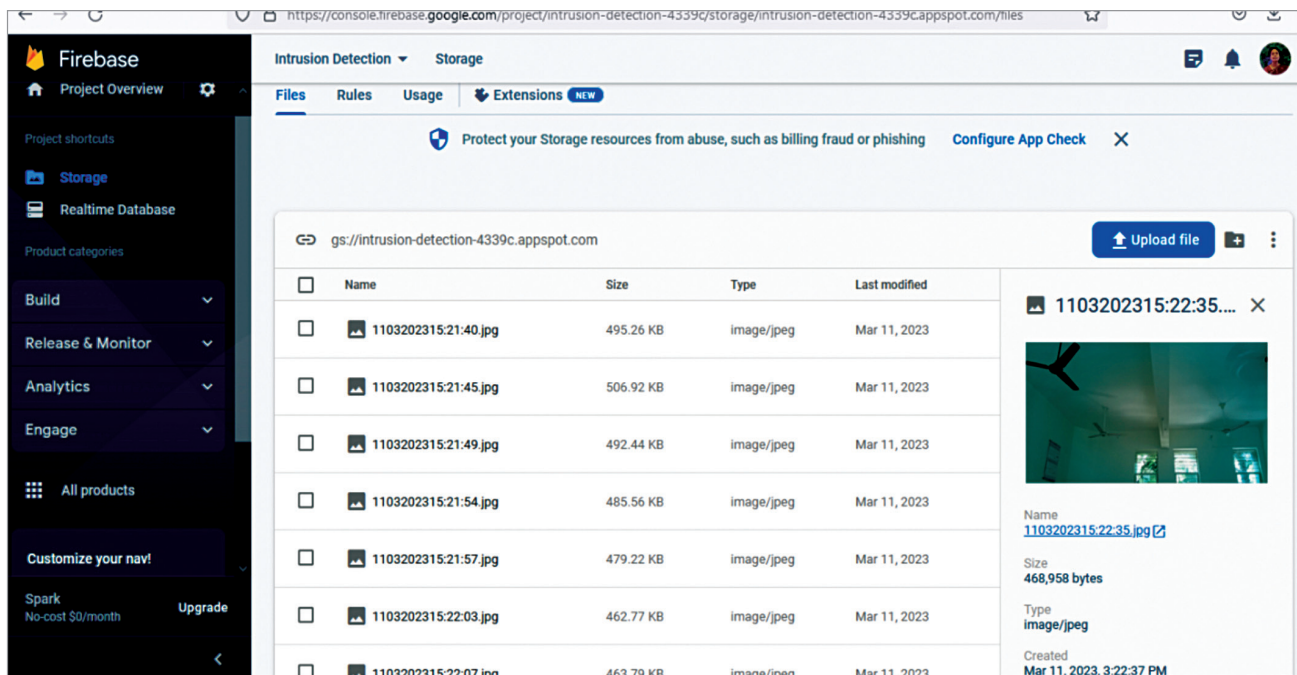


Fig. 8: Author's Firebase storage portal of the observation zone with a Pi camera

South1.) This will lead to creating a default bucket.

6. In Storage, go to Rules, select Edit Rules. We need read and write access in all possible conditions, so in the program on the right hand side window, we have to delete “: if false” and click on the button Publish.

7. Go to Build and select Real Time Database. Click on the button Create Database and opt for Start in test mode. Your data is open by default to enable quick setup. However, you must update your security rules within 30 days to enable long-term client read/write access to ensure Third Party Allowance.

Click on the button Enable. This should provide you with a link to your project database. For example, the link for the author's project was: <https://console.firebase.google.com/project/intrusion-detection-4339c/database/intrusion-detection-4339c-default-rtdb/data/~2F>

8. Go to Project Overview. This will add an app to get started. You will have a choice to select from Android, iOS, and Web App. (The au-

thor opted for Web App.) To ensure a web based access to the project details, select the button Web.

9. Register the app by entering the name of Project Intrusion Detection in the space to enter App Nickname. Select the button Register App.

10. Go to the button Add Firebase SDK and select the button Use npm (<https://www.npmjs.com>)

We now need authentication details in terms of JS SDK, that is your web app's Firebase configuration apiKey, which must include the following details:

authDomain:
databaseURL:
projectId:
storageBucket:
messagingSenderId:
appId:
measurementId:

Remember these details are unique to you and be careful about sharing them with anyone. For Firebase JS SDK v7.20.0 and later, measurementId is optional.

These details are to be included in the Raspberry Pi program for authentication and further processing. Click on Continue to Console. Fig. 6 shows screenshots of Step 5 onwards of your project.

Now create the Python program where you need to fill in the credentials of Firebase you noted earlier. Create the while loop to check the PIR motion sensor. When motion is detected, it will store the picture in Firebase. Fig. 7 shows the code's snippet to set Firebase credentials.

If all the above-mentioned steps are taken correctly, you should be able to obtain all the PIR sensed images, as shown in Fig. 8. **EFY**

EFY Note

The source code of this project is available for free download at www.electronicsforu.com

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